

POSTER ABSTRACT: Australian Virology Group Meeting 2009

Molluscum contagiosum virus detection by real-time PCR

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Molluscum contagiosum virus (MCV) is a member of the poxvirus family, which causes skin lesions. MCV and variola are the only pox viruses that solely infect humans. It is a double-stranded DNA virus of approximately 180kpb in size, and is distributed worldwide. MCV generally remains localized at the site of the lesions; however immuno-compromised people, such as HIV infection, may have lesions that persist for many months to years. Diagnosis is commonly based on the appearance of the lesions; however a differential diagnosis may be required.

Detection at Forensic and Scientific Services since 2004 has relied on a conventional single round gel-based PCR assay. A rapid real-time TaqMan assay for MCV has been developed to aid in the diagnosis of this virus. Probes and primers were designed using sequences of the p43k gene available on Genbank. Sequencing primers were also designed to cover a 303bp region over-spanning the TaqMan target region. The sensitivity of this test is 100% compared to 85% for our previous PCR test. The assay allows rapid turn around of results, and if necessary a high-throughput.